

Bing Li

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Postdoctoral Researcher | CFAR, A*STAR

EDUCATION

Beihang University Computer Application Technology (Advisor: Prof. Di Huang)	Sep 2018 - Jun 2024 Beijing, China
Northeastern University (Master) Signal and Information Processing	Sep 2013 - Jun 2015 Shenyang, LiaoNing Province, China
Northeastern University (Bachelor) Electronic Information Engineering	Sep 2009 - Jun 2013 Shenyang, LiaoNing Province, China

Employment

CFAR, A*STAR, Singapore Postdoctoral Researcher (Advisor: Dr. Guo Qing)	Oct 2024 - Present Research in Adversarial Attacks for Autonomous Driving Systems
China Mobile, Shenyang, China Staff Engineer	Jun 2015 - Sep 2018 Network infrastructure development and maintenance

Research Interests

- Computer Vision, Image/Video Understanding
- 3D Generation, Adversarial Attacks, Gaussian Splatting

Publications

- [1] **Bing Li**, Wuqi Wang, Yanan Zhang, et al. OBJVanish: Physically Realizable Text-to-3D Adversarial Generation of LiDAR-Invisible Objects. //Conference on Neural Information Processing Systems (submitted to ICLR 2025).
- [2] **Bing Li**, Zhijun Chen, Jingzheng Li, et al. CompSOT: Single-Object Tracking in Compressed Video via Progressive Multi-Modal Matching (submitted to IEEE Transactions on MultiMedia 2025).
- [3] **Bing Li**, Guohao Li, Jiaxin Chen, Dongming Zhang, Xiuguo Bao, Di Huang. Cross-modal Contrastive MAE for Compressed Video Pre-training. (TIP 2025).
- [4] **Bing Li**, Jiaxin Chen, Xiuguo Bao, Di Huang. Compressed Video Prompt Tuning. //Conference on Neural Information Processing Systems (NeurIPS 2023).
- [5] **Bing Li**, Jiaxin Chen, Dongming Zhang, Xiuguo Bao, Di Huang. Representation Learning for Compressed Video Action Recognition via Attentive Cross-modal Interaction with Motion Enhancement. //International Joint Conference on Artificial Intelligence (IJCAI 2022).
- [6] **Bing Li**, Longteng Kong, Dongming Zhang, Xiuguo Bao, Di Huang, YunHong Wang. Towards practical compressed video action recognition: a temporal enhanced multi-stream network. //International Conference on Pattern Recognition (ICPR 2020).
- [7] Xueping Wang, Xueni Guo, Feihu Yan, **Bing Li**, Guangzhe Zhao. Progressive Densification of 3D Gaussians for High-Fidelity Talking Head Synthesis (Frontiers of Computer Science 2025).
- [8] Zhewen Zhou, **Bing Li***, Qing Guo, Chenlei Li, Guangshuai Gao. PACTFormer: Peak-Aware Cross-Temporal Transformer for Temporal Action Detection (PRCV 2025).
- [9] Hong Wen, **Bing Li**, Liang Qian, Haoyuan Li, Weiwei Fang. EMM-CVT: An Efficient Multi-Modal Video Transformer for Compressed Video Action Recognition (submitted to AAAI 2026).
- [10] Yanan Zhang, Junhao Xu, Jianan Zhang, **Bing Li**, Yang Zhao, Wei jia. BLC-CP: Breaking Through Limited Communication for Multi-agent Collaborative Perception (submitted to AAAI 2026).
- [11] Xiangtao Zhang, Hailong Yan, Obed Irihose, Joey Tianyi Zhou, Chee Seng Chan, **Bing Li**, Ce Zhu, Le Zhang. Rethinking Federated LLM Fine-Tuning: On Data Quality and Subspace Discrepancy. (submitted to AAAI 2026).
- [12] Jingzheng Li, Tiancheng Wang, Xingyu Peng, Jiacheng Chen, Zhijun Chen, **Bing Li**, Xianglong Liu. Safety2Drive: Safety-Critical Scenario Benchmark for the Evaluation of Autonomous Driving (submitted to AAAI 2026).
- [13] Mengkun Guo, Die Tao, **Bing Li***, Bohao Qu, Qing Guo, Ming Ma. Adapting VLMs for Compressed Video Action Recognition via Multimodal Prompt Tuning (submitted to AAAI 2026).
- [14] Jingzheng Li, Xianglong Liu, Shikui Wei, Zhijun Chen, **Bing Li**, Qing Guo, Xianqi Yang, Yanjun Pu, Jiakai Wang. Towards Benchmarking and Assessing the Safety and Robustness of Autonomous Driving on Safety-critical Scenarios. (submitted to IEEE Transactions on Information Forensics & Security 2025).
- [15] Yongheng Tang, Shuifa Sun, **Bing Li**, Haoning Wang, Jing Zhang. GSHNet: A Gated and Saliency-guided Hierarchical Network for Power Line Segmentation in aerial images. (submitted to IEEE Transactions on Automation Science and Engineering 2025).
- [16] Dongmei Chen, Guiji Sun, **Bing Li**, Shuhua Liu, Jv Luo, Xianfeng Zhou, Jingcheng Zhang. TSHFS: A Two-Stage Hyperspectral Feature Selection Approach for Congeneric Rice Planthopper Species Classification (submitted to Pest Management Science 2025).
- [17] Yongheng Tang, Shuifa Sun, **Bing Li**, Yiyu Guo, Ben Wang, Qing Guo. ShadowLane: Natural & Adversarial Shadow Threats to Lane Detection. (submitted to IEEE Transactions on MultiMedia 2025).

Projects

Deep hash video retrieval

Jun 2019 - Jun 2020

hybrid GPU/CPU decoding

Classification and annotation of video complex scene datasets

Dec 2018 - Feb 2020

dataset construction and automate labeling

Skills

Machine Learning: video analysis, adversarial learning, image processing

Programming: Python, C++, PyTorch

Academic Services

Reviewer of IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) and IEEE Transactions on MultiMedia (TMM)

Reviewers for international conferences (NeurIPS, IJCAI, AACL, MM, CVPR, and ECCV)